Logit Regression Results

==========

=====

Dep. Variable: low No. Observations: 189

Model: Logit Df Residuals: 185

Method: MLE Df Model: 3

Date: Sun, 25 Oct 2026 Pseudo R-squ.: 0.06250

Time: 23:06:42 Log-Likelihood: -110.00

converged: True LL-Null: -117.34

Covariance Type: nonrobust LLR p-value: 0.002125

==========

=====

coef std err z P>|z| [0.025 0.975]

Intercept -1.3551 0.241 -5.611 0.000 -1.828 -0.882

smoke 0.6831 0.329 2.075 0.038 0.038 1.329

ht 1.4174 0.624 2.273 0.023 0.195 2.640

ui 1.0384 0.428 2.426 0.015 0.200 1.877

==========

=====Confusion Matrix:Predicted 0 1

Actual

0 119 11

1 45 14

1. The model's accuracy is 70.37%, 119 + 14 = 133, divided by the total 189.
2. The model's lowest prediction is 56%, the total predicted positive column 11 + 14 = 25, 14 divided by 25 is 0.56.
3. Both categories have at least 30 support. Class 0 has 130 and Class 1 has 59.
4. The pseudo-R^2 value of 0.06250 is very low, meaning the model is not good in this scenario and only explains a small fraction of the variance.

Poisson Regression Results

==========

=====

Dep. Variable: mhlth_crudeprev No. Observations: 2299

Model: Poisson Df Residuals: 2287

Method: MLE Df Model: 11
Date: Sun, 25 Oct 2026 Pseudo R-squ.: 0.04356
Time: 23:12:23 Log-Likelihood: -5397.6
converged: True LL-Null: -5643.4
Covariance Type: nonrobust LLR p-value: 1.908e-98

	coef	std err	z	P>|z|	[0.025	0.975]
Intercept	1.7032	0.172	9.888	0.000	1.366	2.041
binge_crudeprev	0.0032	0.003	0.954	0.340	-0.003	0.010
checkup_crudeprev	-0.0004	0.002	-0.192	0.848	-0.004	0.004
csmoking_crudeprev	0.0043	0.004	1.100	0.271	-0.003	0.012
depression_crudeprev	0.0137	0.002	5.800	0.000	0.009	0.018
obesity_crudeprev	0.0001	0.002	0.076	0.940	-0.003	0.004
sleep_crudeprev	0.0024	0.002	1.059	0.289	-0.002	0.007
selfcare_crudeprev	-0.0608	0.022	-2.708	0.007	-0.105	-0.017
indeplive_crudeprev	0.0522	0.014	3.759	0.000	0.025	0.079
loneliness_crudeprev	0.0113	0.003	3.717	0.000	0.005	0.017
foodinsecu_crudeprev	-0.0083	0.007	-1.247	0.212	-0.021	0.005
housinsecu_crudeprev	0.0150	0.008	1.786	0.074	-0.001	0.032

Variance Inflation Factor (VIF):

Feature	VIF
const	1.000000
binge_crudeprev	2.528842
checkup_crudeprev	1.750444
csmoking_crudeprev	6.934794
depression_crudeprev	2.216376
obesity_crudeprev	2.573017
sleep_crudeprev	3.209334
selfcare_crudeprev	30.514413
indeplive_crudeprev	30.953309
loneliness_crudeprev	3.432565
foodinsecu_crudeprev	62.830787
housinsecu_crudeprev	46.839954

1. The four significant predictors are depression_crudeprev, selfcare_crudeprev, indeplive_crudeprev, and loneliness_crudeprev where the p-value is less than 0.05.
2. Selfcare_crudeprev is the most significant, it has the largest coefficient magnitude at -0.0608.
3. Both selfcare_crudeprev (30.51) and indeplive_crudeprev (30.95) have an overinflated VIF.
4. About 4.36% of the variability is captured by the model (Pseudo R-squared = 0.04356).